Modeling and Simulation of a Campus Evacuation as a Diffusion Process

Carleton University

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1. Introduction

- City transportation systems: reduce **congestion**, **CO2 emissions**, **travel time**
- Planners: advanced **urban planning tools** to model, simulate, evaluate strategies

2. Background

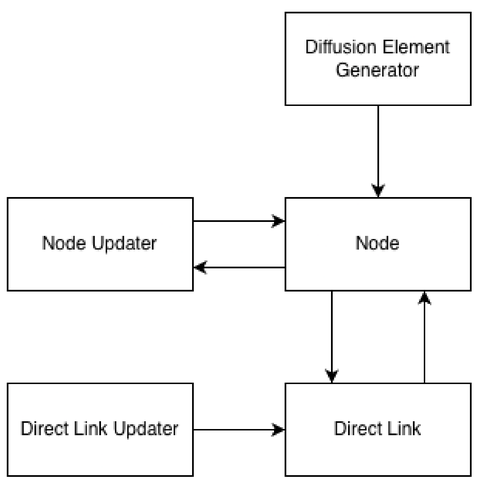

Fig 1. Diffusion Abstract Model (DAM)

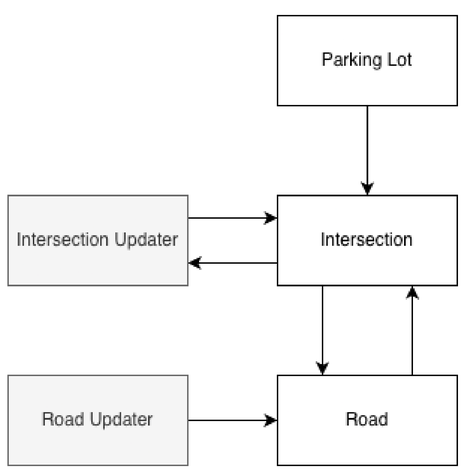

Fig 2. DAM for a transportation network of Parking Lots, Intersections, and Roads

3. System Architecture and Design

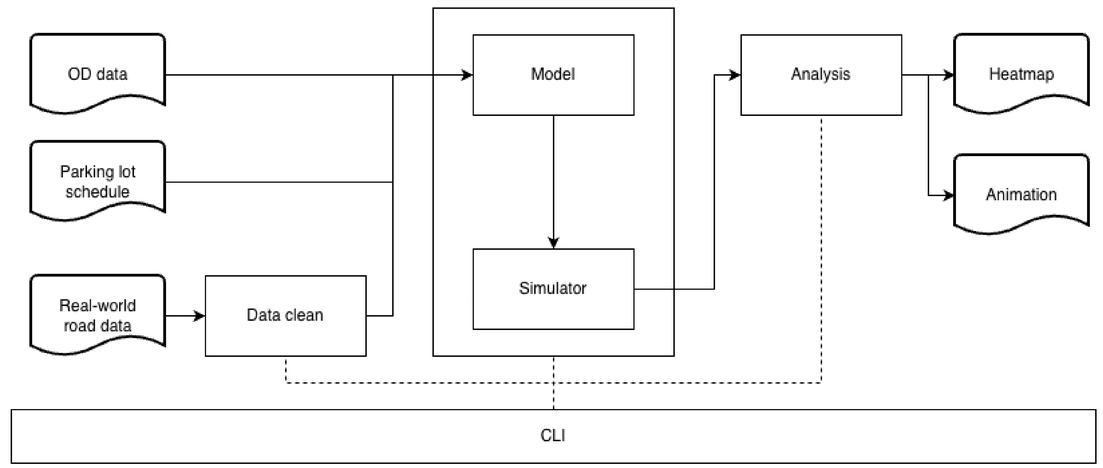

Fig 3. OD data, parking lot schedules, real-world road data to configure the model. Simulation's results: heatmap and animation of the simulation. The system integrated with a CLI.

- Architectural characteristics: **configurability** and **reusability**
- Configure agent behaviour with **origin-destination (OD) data**
- Design and implement models using the **Discrete Event System Specification (DEVS)**

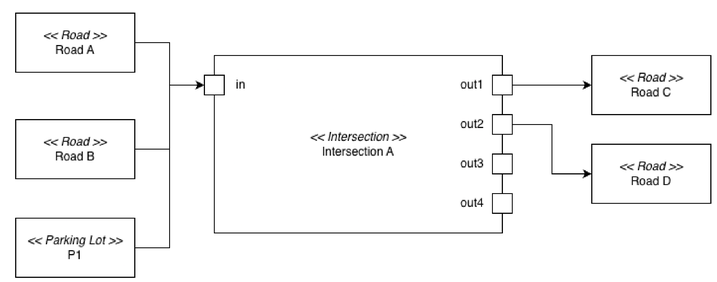

Fig 4. DEVS coupled model: Intersection A receives traffic from Roads and Parking Lots, then routes traffic to coupled Roads based on its OD data.

4. Campus Evacuation Case Study

Experimental frame:

- Parking lots at **maximum capacity**
- Vehicles **periodically** exit Parking Lots
- Ignore pedestrians, cyclists, buses
- Congestion as **emergent behaviour**

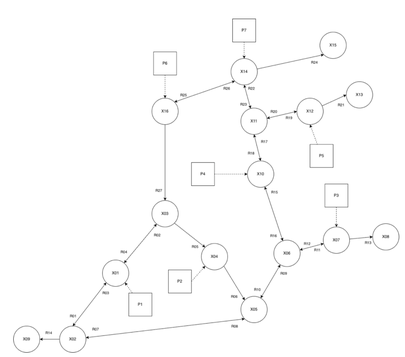

Fig 5. Network diagram of Carleton University's campus.

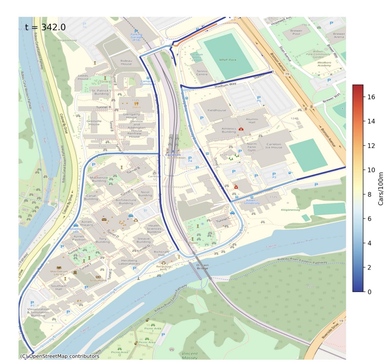

Fig 6. Animation frame of a simulation run.

5. Results Analysis

- Iteratively ran and evaluated experiments to converge on a solution.
- Initial scenario (all parking lots evacuate immediately): congestion at multiple exits.
- Scenario with **minimal congestion:** delay P3 and P4 by 1 hour and P6 by 1.5 hours.

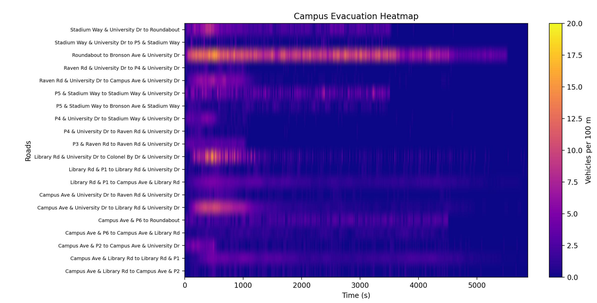

Fig 7. Scenario 1 heatmap. All parking lots evacuate immediately.

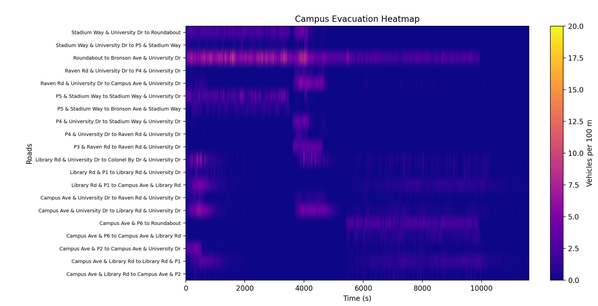

Fig 8. Scenario 6. Parking lots 3 and 4 delayed by 1 hour, parking lot 6 delayed by 1.5 hours.

6. Project GitHub Repository